

HDOC DAY – 26

Aim

To write a C program that displays a menu to print different right-aligned patterns using nested loops, functions, and a switch statement.

Algorithm

1. Start

2. Define Functions:

a. printTriangle(int n):

- i. For row i from 1 to n:
 1. Print n - i spaces.
 2. Print i stars (*).
 3. Move to the next line.

b. printNumberTriangle(int n):

- i. For row i from 1 to n:
 1. Print n - i spaces.
 2. Print numbers from 1 to i.
 3. Move to the next line.

c. printInvertedTriangle(int n):

- i. For row i from n down to 1:
 1. Print n - i spaces.
 2. Print i stars (*).
 3. Move to the next line.

3. Main Function:

- a. Display menu options:
 - i. Right-aligned increasing triangle
 - ii. Right-aligned increasing number triangle
 - iii. Right-aligned decreasing triangle
- b. Take choice from the user.
- c. Take the number of rows (n) from the user.

- d. Use a switch statement to call the corresponding function based on choice.

4. Stop

C CODE :

```
#include <stdio.h>
```

```
// 1. Function for Right-Aligned Star Triangle
```

```
void pattern1(int n) {  
    int i, j;  
    for (i = 1; i <= n; i++) {  
        // print spaces  
        for (j = 1; j <= n - i; j++) {  
            printf(" ");  
        }  
        // print stars  
        for (j = 1; j <= i; j++) {  
            printf("*");  
        }  
        printf("\n");  
    }  
}
```

```
// 2. Function for Right-Aligned Number Triangle
```

```
void pattern2(int n) {  
    int i, j;  
    for (i = 1; i <= n; i++) {  
        // print spaces  
        for (j = 1; j <= n - i; j++) {  
            printf(" ");  
        }  
        // print numbers  
        for (j = 1; j <= i; j++) {  
            printf("%d", j);  
        }  
        printf("\n");  
    }  
}
```

// 3. Function for Right-Aligned Decreasing Triangle

```
void pattern3(int n) {  
    int i, j;  
    for (i = 1; i <= n; i++) {  
        // print spaces  
        for (j = 1; j < i; j++) {  
            printf(" ");  
        }  
    }  
}
```

```
        // print stars
        for (j = 1; j <= n - i + 1; j++) {
            printf("*");
        }
        printf("\n");
    }
}
```

```
int main() {
    int choice, n;

    printf("1. Right aligned star triangle\n");
    printf("2. Right aligned number triangle\n");
    printf("3. Right aligned decreasing triangle\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    printf("Enter number of rows: ");
    scanf("%d", &n);

    switch (choice) {
        case 1:
            pattern1(n);
            break;
```

case 2:

pattern2(n);

break;

case 3:

pattern3(n);

break;

default:

printf("Wrong choice!\n");

}

return 0;

}

CODE SCREENSHOT :

```
UW PICO 5.09 File: 26q.c

#include <stdio.h>

// 1. Function for Right-Aligned Star Triangle
void pattern1(int n) {
    int i, j;
    for (i = 1; i <= n; i++) {
        // print spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print stars
        for (j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// 2. Function for Right-Aligned Number Triangle
void pattern2(int n) {
    int i, j;
    for (i = 1; i <= n; i++) {
        // print spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print numbers
        for (j = 1; j <= i; j++) {
            printf("%d", j);
        }
        printf("\n");
    }
}

// 3. Function for Right-Aligned Decreasing Triangle
void pattern3(int n) {
    int i, j;
    for (i = 1; i <= n; i++) {
        // print spaces
        for (j = 1; j < i; j++) {
            printf(" ");
        }
        // print stars
        for (j = 1; j <= n - i + 1; j++) {
            printf("*");
        }
        printf("\n");
    }
}

int main() {
    int choice, n;

    printf("1. Right aligned star triangle\n");
    printf("2. Right aligned number triangle\n");
```

```
printf("3. Right aligned decreasing triangle\n");
printf("Enter choice: ");
scanf("%d", &choice);

printf("Enter number of rows: ");
scanf("%d", &n);

switch (choice) {
    case 1:
        pattern1(n);
        break;
    case 2:
        pattern2(n);
        break;
    case 3:
        pattern3(n);
        break;
    default:
        printf("Wrong choice!\n");
}

return 0;
}
```

Output screenshot :

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % gcc 26q.c -o 26q
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./26q
1. Right aligned star triangle
2. Right aligned number triangle
3. Right aligned decreasing triangle
Enter choice: 1
Enter number of rows: 5
  *
 **
***
****
*****
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./26q
1. Right aligned star triangle
2. Right aligned number triangle
3. Right aligned decreasing triangle
Enter choice: 2
Enter number of rows: 5
  1
 12
123
1234
12345
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./26q
1. Right aligned star triangle
2. Right aligned number triangle
3. Right aligned decreasing triangle
Enter choice: 3
Enter number of rows: 5
*****
****
***
**
*
```